

Erdős Problem #584

EP-584	L1 - Tractable	Partiellement résolu	Graph Theory
--------	----------------	----------------------	--------------

Énoncé original (anglais, reproduction mot pour mot)

Let G be a graph with n vertices and δn^2 edges. Are there subgraphs

$H_1, H_2 \subseteq G$ such that

$\{$

$\{$

- H_1 has $\gg \delta^3 n^2$ edges and every two edges in H_1 are contained in a cycle of length at most 6 , and furthermore if two edges share a vertex they are on a cycle of length 4 , and

$\{$

- H_2 has $\gg \delta^2 n^2$ edges and every two edges in H_2 are contained in a cycle of length at most 8 .

$\}$

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>